Data Science and Analytics

“Environmental Factors that Impact College Admissions”

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Introduction

It is common knowledge that college in America is difficult to get into, as high school students spend the majority of their time trying to build up their resume in order to better their chances of getting into a prestigious institution.

According to the College Board, the factors that most affect admission decisions are:

- Grades received & Standardized test scores
- Personal statements, essays, and recommendations
- Extracurricular activities

However this issue raises many questions regarding how much of the college admissions process do students actually have control over. In Grutter v. Bollinger, a case from 2003, the Supreme Court approved the use of race as a factor in college admission in order to produce a diverse student body, proving that there are environmental factors that impact the college admissions process as well. With affirmative action being reconsidered in the US Supreme Court, the future of college admissions seems uncertain. At the University of Michigan, for example, Black undergraduate admission declined from 7 percent in 2006 to 4 percent in 2021, even though the total percentage of college-age African Americans in Michigan increased from 16 percent to 19 percent.

An analysis of the environmental factors surrounding college admissions is crucial in understanding the ethics of college admissions as well as how much control students even have over their chance of admission.

Data Dictionary

Environmental Factors: features that arise from the surroundings and are not under the control of the applicant

Race: an ethnic group (races examined were white, asian, black, hispanic/latino, indigenous, and other - groupings based on official university demographics)

- Affirmative Action: sets of policies and practices within a government or organization seeking to include particular groups based on their gender, race, sexuality, creed or nationality in areas in which such groups are underrepresented
- Admitted Student Body: student population that was accepted into the school
- Applicant Pool: total population of students who applied to the school

Geographical Diversity: the amount of different geographical origins present in a student body

Legacy Status: preference given by an institution or organization to certain applicants on the basis of their familial relationship to alumni of that institution

Machine Learning: a branch of artificial intelligence (AI) and computer science which focuses on the use of data and algorithms to imitate the way that humans learn, gradually improving its accuracy

- Bins: simplified and compressed columns of data with reduced number of possible values or levels
- Algorithm: finite sequence of rigorous instructions, typically used to solve a class of specific problems or to perform a computation
- Lift Chart: graphical representation of the improvement that a mining model provides when compared against a random guess, and measures the change in terms of a lift score

Purpose

The research in this project was focused on the major environmental factors that impact college admissions in order to learn whether or not the college admissions process as it stands is even ethical and to predict what the future of college admissions will look like if the current trends continue. By understanding this we can further the conversation on the power these large institutions have as they design these admission systems to serve themselves.

The major environmental factors examined were the applicant's:

- Race
- Location
- Status as a legacy

The research examined the influence these factors have on admissions at universities across the United States and how the data regarding the students admitted matched up to the data of the total applicant pool.

Methods

My first method to obtain data is through the use of documents and records. This is an efficient method to use as it is pre-existing data. By observing the change over time, I can make a hypothesis of how it will be in the future and get a view of how much needs to be revised.

I collected data from the following universities/sources:

- Stanford University
- Harvard University
- University of Pennsylvania
- U.S. National Center for Education Statistics

Another vital method for my data obtention was Correlation Research. This specific type of research is non-experimental so it aligns with my chosen topic. It is often measured with two variables (in my case the given environmental factor and makeup of the admitted student body) and assesses the relationship between the two. This will allow me to determine if one of the variables directly affects the other and vice versa.

Once I had the data I used data cleaning in order to separate out the data I needed to create the visual representations and ensure that it was ready for analysis. In order to analyze the data I utilized machine learning with code and the AWS Sagemaker and DataRobot platforms. The algorithms attached to these platforms are much more powerful and accurate than algorithms that can be derived manually, and are essential in ensuring my analysis is as accurate as possible. The world of data science and machine learning heavily depend on each-other and this project proved that.

Results

Race

Table 1. Racial Demographics of the *Admitted Undergraduate Class*, Fall 2022

University	White	Asian	Black	Hispanic/ Latino	Indigenous	Other
Stanford	26%	26%	7%	18%	1%	22%
Harvard	40%	14%	15%	13%	3%	15%
UPenn	31%	26%	8%	13%	1%	21%
Average	34%	22%	10%	15%	2%	17%

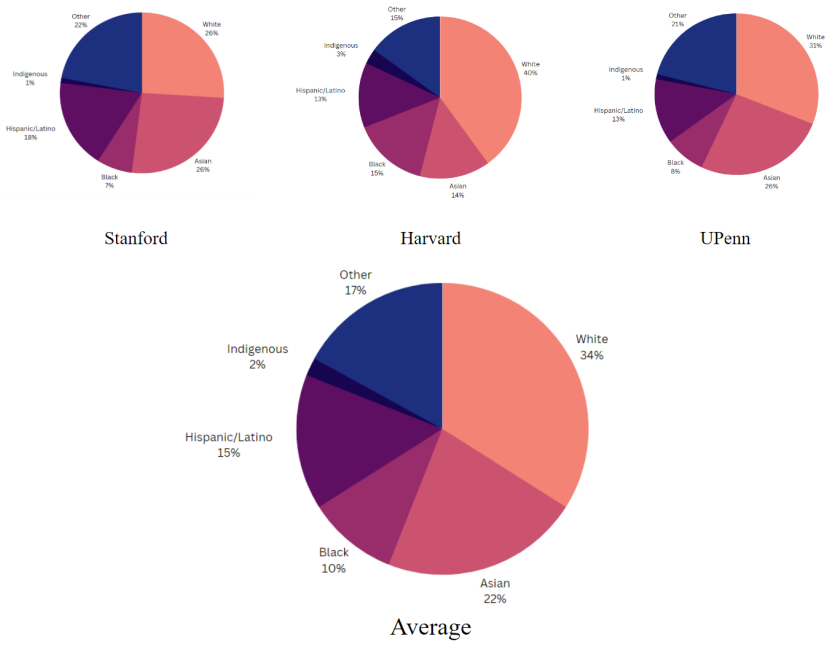


Figure 1. Pie Chart of *Individual and Average Demographics of Admitted Clas*s, Fall 2022

Looking at the visual representations of the data above, there is pretty minimal trend, with more similarity between UPenn and Stanford. White and Asian make up the majority in all three.

Table 2. Racial Demographics of the *Applicant Pool*, Fall 2022

University	White	Asian	Black	Hispanic/ Latino	Indigenous	Other
Stanford	34%	32%	4%	12%	1%	17%
Harvard	47%	26%	7%	8%	1%	11%
UPenn	41%	32%	6%	8%	1%	12%
Average	41%	30%	5%	9%	1%	14%

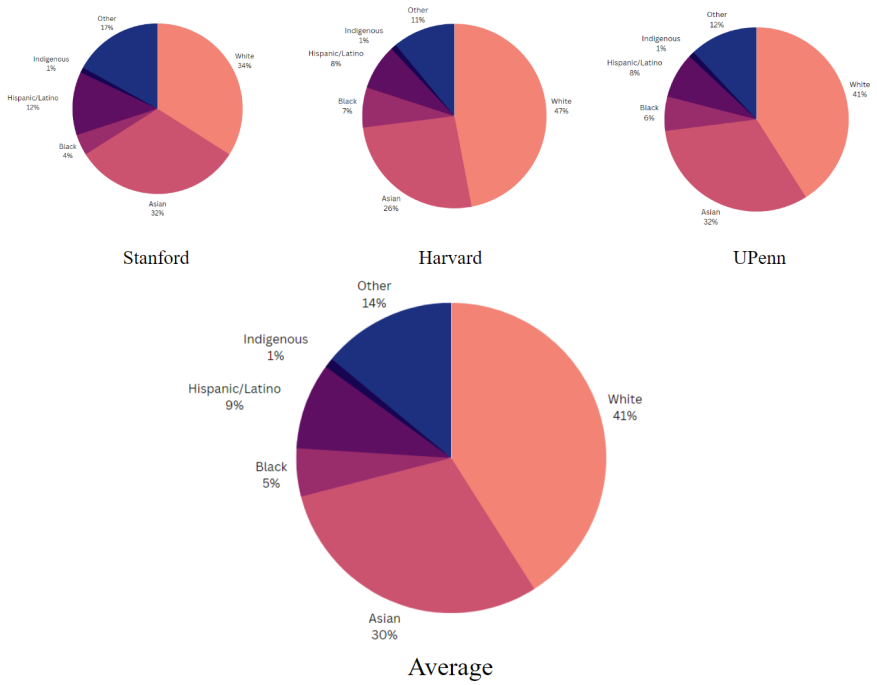


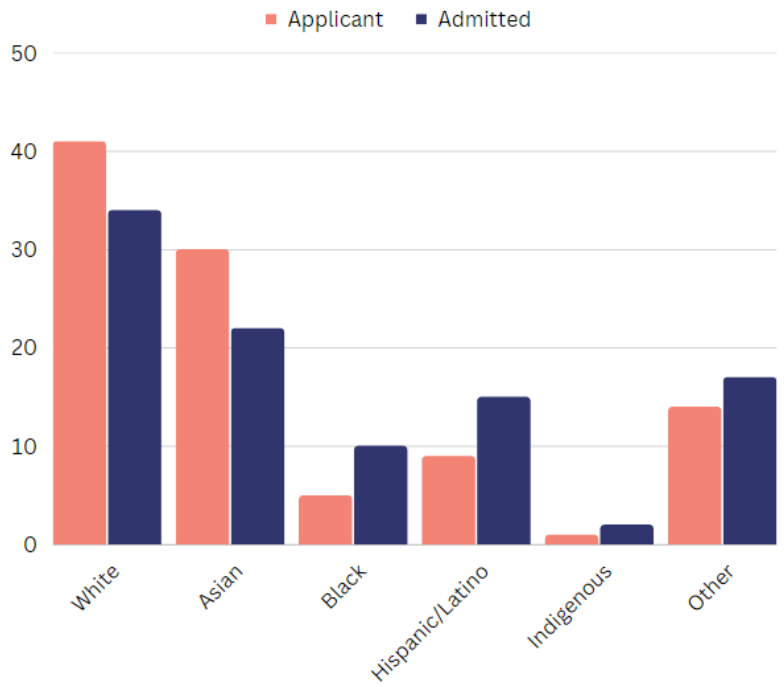
Figure 2. Pie Chart of *Individual and Average Demographics of Applicant Pool*, Fall 2022

Similar to the previous data set there is also minimal trend in this set of data, this time with Harvard and UPenn sharing the most similarity. White and Asian are an even greater majority, and this will be important to note as we dive deeper into the analysis.

Table 3. Contrasts Between the *Averages of Both Datasets*

Race	Difference	Percent Change
White	-7	-18%
Asian	-8	-27%
Black	+5	+100%
Hispanic/Latino	+6	+66%
Indigenous	+1	+100%
Other	+3	+21%

Figure 3. Bar Graph of Average Demographics of the *Applicant Pool vs. Admitted Class*



This data is very revealing as it shows that the demographics of the total applicant pool do not match up with the demographics of the admitted class, which is what should have been the case if the admissions process was truly color-blind (does not consider race). As shown in the table above, the difference is very great, ranging from -18% to +100%.

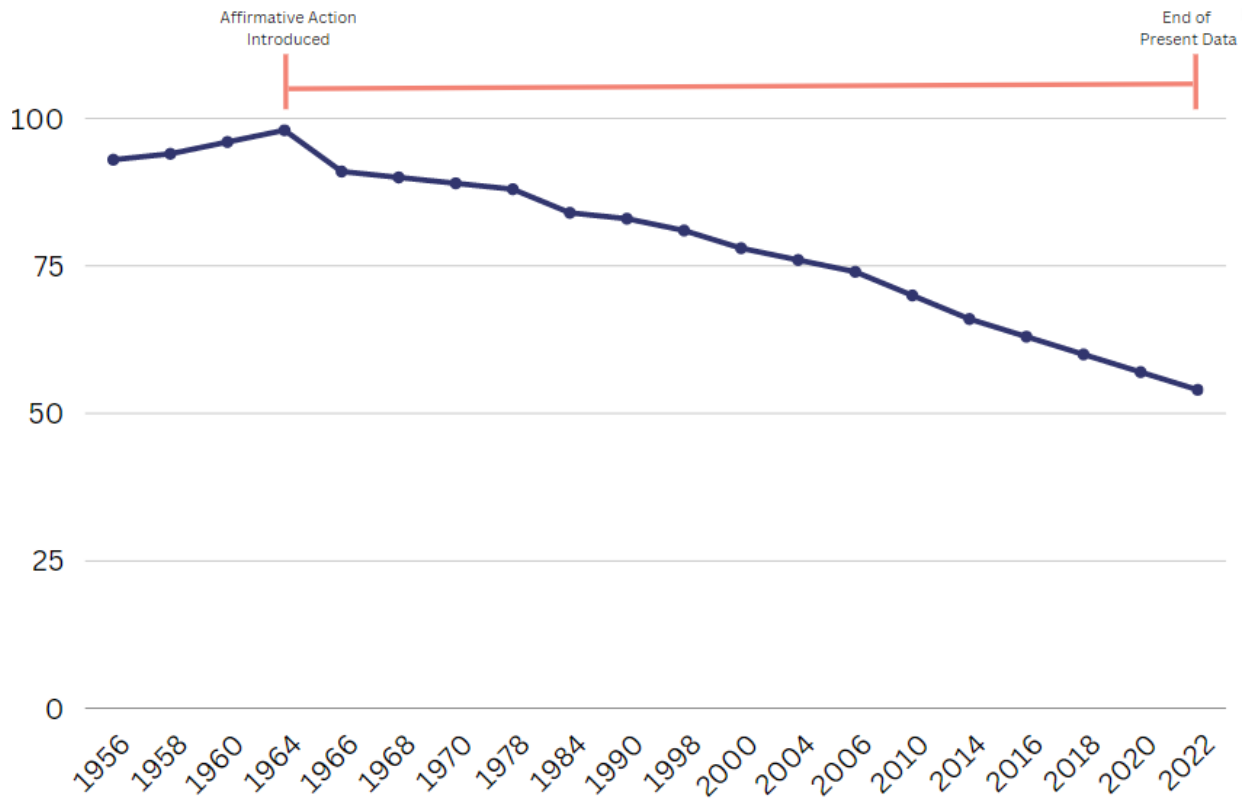
Table 4. Percentage of *White Students Admitted to an American University*, 1956-2022

Year	Percent White	Affirmative Action[1]
1956	93	1
1958	94	1
1960	96	1
1964	98	1
1966	91	0
1968	90	0
1970	89	0
1978	88	0
1984	84	0
1990	83	0
1998	81	0
2000	78	0
2004	76	0
2006	74	0
2010	70	0
2014	66	0
2016	63	0
2018	60	0
2020	57	0
2022	54	0

This data shows a clear difference once affirmative action was introduced, as the percentage of white students began to rapidly decrease compared to its previous increase in the 1950s - 1960s.

[1]Assigned a binary value so the machine learning program can read it as an integer (0 - yes 1- no)

Figure 4. Visual Representation of the Data w/ Timeline, 1956-2022

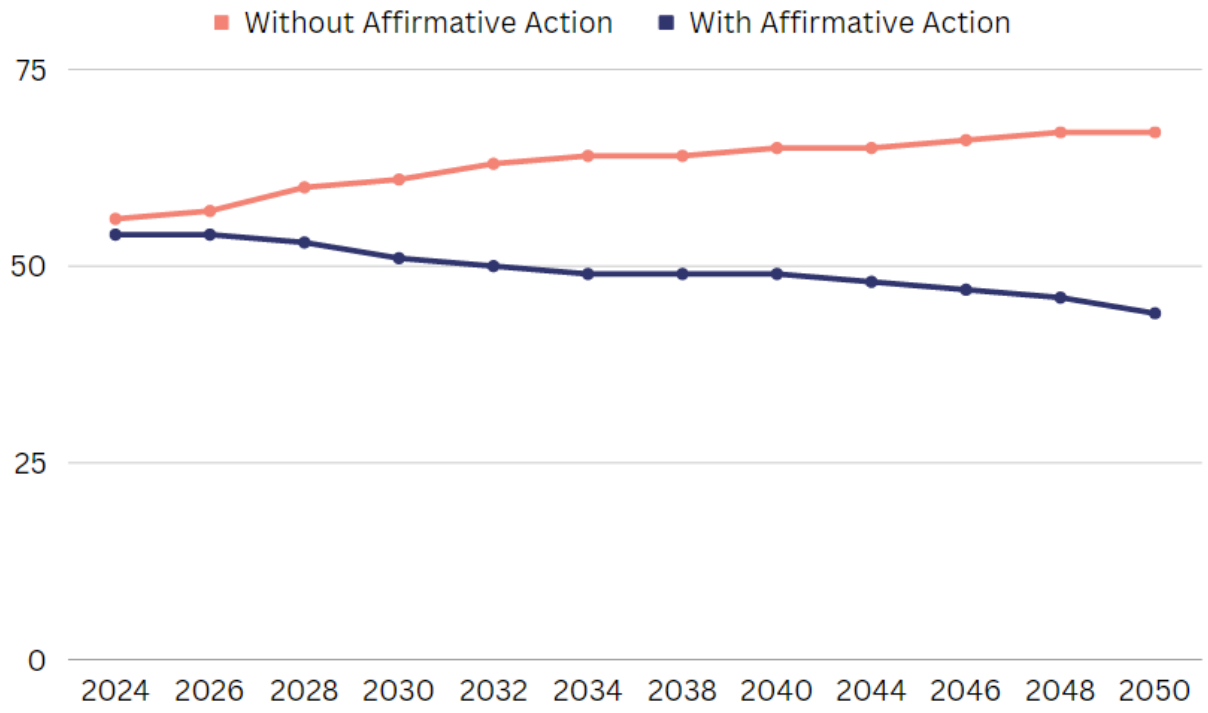


Now the data is more visually clear, and when presented against the timeline it becomes even more apparent that the introduction of affirmative action changed the trend of the data.

I want to test what the predictions will look like with and without affirmative action, to see if the algorithm thinks it has an impact.

To do this, I need to create 2 data sets: one without affirmative action and one with to plug into the Machine Learning Model once it has been created (done in AWS Sagemaker & DataRobot with open source code and algorithms)

Figure 5. Predictions Made by the Model *With and Without Affirmative Action*, 2024-2050

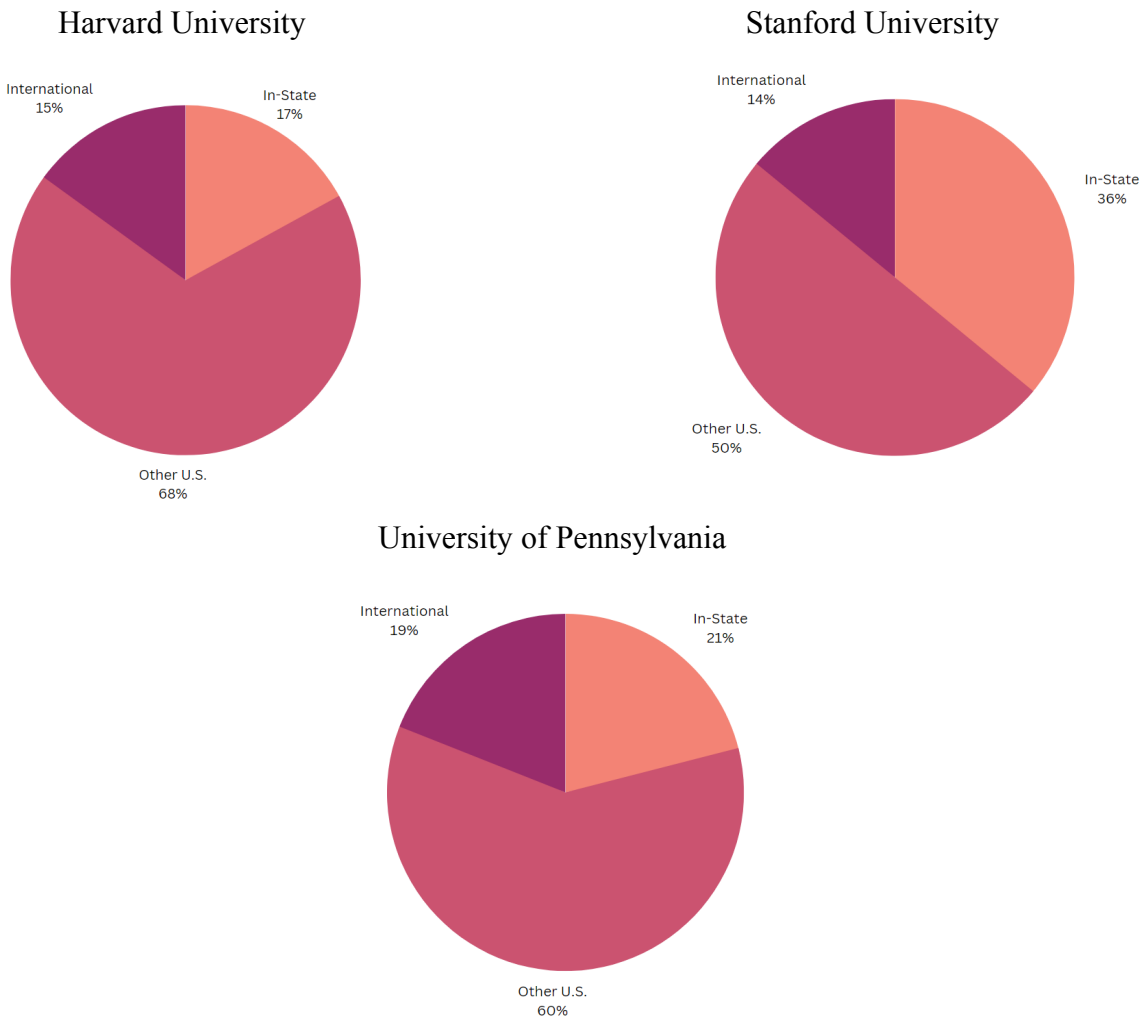


Although the data is similar to start, there is a clear divergence of the two as predicted by the model. This agrees with the previous model, which also saw a shift in trend once affirmative action was introduced.

- With affirmative action, the percentage of white students decreased, meaning that more ethnically diverse students would be accepted
- Without affirmative action, the percentage of white students increased, contrasting the trend that affirmative action created

Location

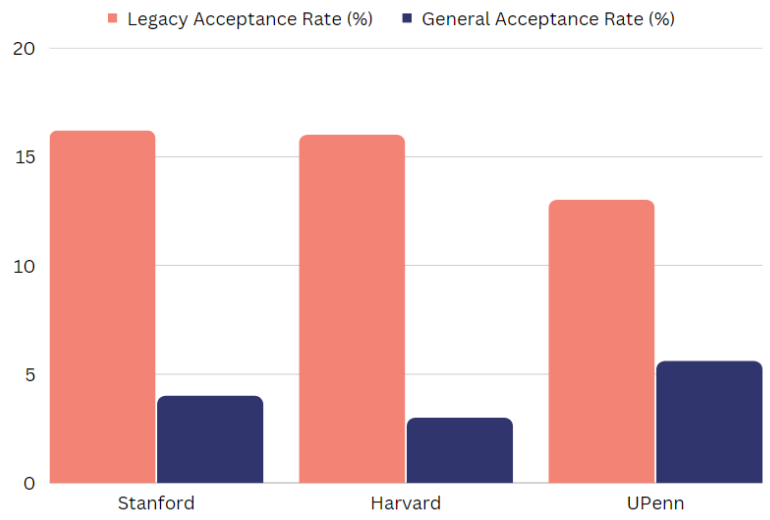
Figure 6. Pie Chart of *Geographic Diversity of Accepted Undergraduate Class*, Fall 2022



The data does differ a bit between the three universities examined, however not so much that it warrants analyzing any correlation. What this does show us is that students who are in-state have the highest chance of being admitted, seeing as they make up a large percentage of the total admitted body when compared to the other 49 states and foreign countries. It is important to note that similar to race, universities are looking to expand upon their geographic diversity for marketing purposes and according to researchers at the Harvard Crimson, “colleges will almost always favor the applicant coming from a more distant or exotic locale” (Fahs & Lewis, 2015).

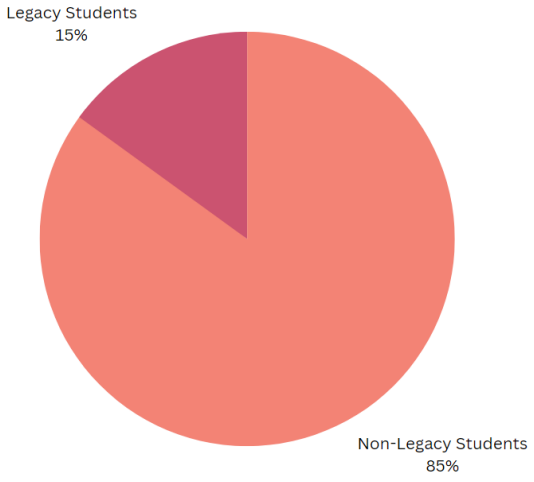
Status as a Legacy

Figure 7. Bar Graph Comparison of *Legacy Student vs General Acceptance Rate*, Fall 2022



Looking at this alone, it clearly shows that legacies have an advantage when it comes to admissions in comparison to non-legacy students. However the chart below shows that legacy students do not make up the majority of the student body, because the lower general acceptance rate applies to a larger population than the legacy acceptance rate does .

Figure 8. Pie Chart of *Average Percentage in Student Body*, Fall 2022



Conclusions

Race

In general the three universities examined saw very similar data in the demographics of both their admitted body and applicant pool, so we can conclude that the prestigious universities like to keep their racial demographics at around the same level. What change we did see was from the applicant pool to the admitted class, as the visual graph showed that certain demographics have almost doubled whilst others have decreased by almost 30%. The races considered as 'majorities' (white and asian) saw a lower percentage in the admitted class than they did in the applicant pool, while black, indigenous, hispanic/latino, and additional minorities saw an increase in the admitted class.

However if we compare both majorities, Asians still had the greatest decrease in representation in the admitted class. This aligns with what researchers at Princeton University found, in that applicants who identify as Asian need to score 140 points higher on the SAT than a white applicant in order to have an equal chance at admission into an elite college—they dubbed this the "Asian tax" in college admissions (Nimetz 1).

This leads into the next conclusion that we came to, in that affirmative action is preventing the percentage of white students in universities compared to other racial groups from increasing. The past trend in data showed that affirmative action decreased the percent white over the span of decades, and the machine learning regression algorithm predicted that without affirmative action, although not as steep as before, the percent white would still increase. These rationales are useful when considering the ethics of affirmative action, and it proves that race is a factor in college admissions.

Location

Across the three universities examined, they all shared very similar demographics when it comes to the geographical spread of the admitted student body. In all three the largest majority was America, suggesting that American universities accept more American students than international students. Even within America, they are most likely to accept in-state students as they make up the largest percentage, since the percentage for Other U.S. applies to 49 different states.

That being said, while geography does seem to be a minor factor in college admissions, universities are always looking to reach their set diversity 'quotas' for marketing purposes, and may choose applicants to further their geographical diversity, as concluded by Kate Eitchorn, researcher at *The New School*, in her paper on discrimination against students from large cities (Eitchorn 1).

Status as a Legacy

In the visual comparison of the two it is very clear that legacies have a much higher acceptance rate than general applicants, as their acceptance rate is more than double the general acceptance rate, even triple for Harvard and Stanford. This is typical practice at such prestigious universities, seeing as we saw a similar spread across the three. Despite legacies having a higher admission rate, they do not make up a larger portion of the admitted student body, suggesting that there are less legacies in the applicant pool as a whole.

Next Steps

Whether or not this practice is ethical is still up for debate, seeing as many of the factors were made to promote diversity and inclusion, however the universities could just be acting in their best financial interests and marketability,

To further analyze the data regarding race, the future trend with and without affirmative action should include the trend for individual racial groups, as well as examining the trend in other universities. If we analyze the impact of affirmative action on individual racial groups, we can identify who affirmative action is benefitting the most and who it is harming. Just because the percent white increased without affirmative action, it does not mean that the percent for every other minority will decrease because the two may not be correlated in that way.

For both location and status as a legacy, more research should be done regarding the factors that impact these demographics. Would certain global events impact the amount the university accepts from certain groups? Events such as the Ukraine War, various college admissions scandals, and global pandemics could all have an impact on the demographics of the admitted student body as well as the general applicant pool.

Also worth investigating is the crossover between all three of these environmental factors. As an example, many researchers have argued that eliminating legacy preferences is one way to help achieve racial diversity without using affirmative action, which the data shows is discriminatory (Rim 1).

College admissions is very controversial as to whether there even is a problem or not, seeing as there are so many perspectives on the issue. By further understanding the context behind these factors and the impact they have we can resolve the ongoing debate.

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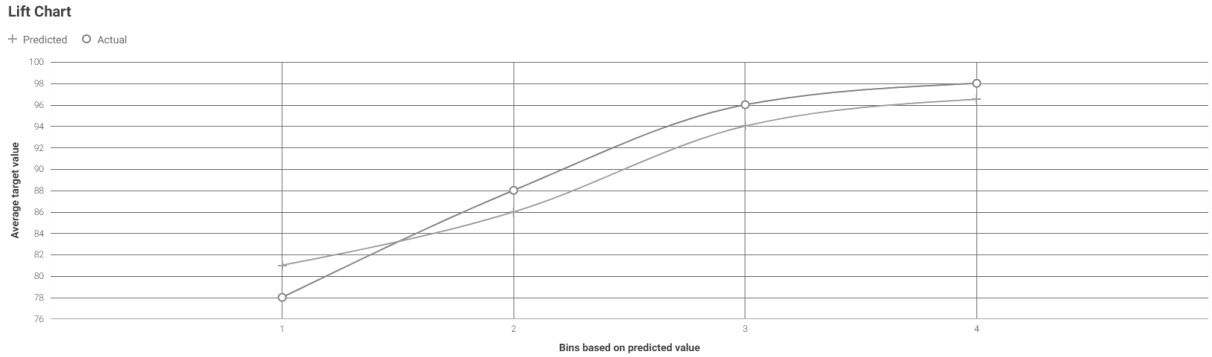
Appendix

The model was trained using the data and determined that the following algorithm was best:

- eXtreme Gradient Boosted Trees Regressor

```
library(xgboost)
data(agaricus.train, package='xgboost')
data(agaricus.test, package='xgboost')
train <- agaricus.train
test <- agaricus.test
bst <- xgboost(data = train$data, label = train$label, max_depth = 2, eta = 1,
               nrounds = 2, objective = "binary:logistic")

## [1] train-logloss:0.233376
## [2] train-logloss:0.136658

xgb.save(bst, 'model.save')

## [1] TRUE

bst = xgb.load('model.save')
pred <- predict(bst, test$data)
```

- The model looks something like this, and supports various objective functions, including regression, classification, and ranking – we utilized its regression ability

Figure 9. Lift Chart of the Machine Learning Model comparing *Bins vs. Average Target Value*

- The lift chart shows the accuracy of the model
- Based on how close the predicted line and actual lines are we can tell that the model is fairly accurate and will serve its purpose